

Diffusion Models in Generative AI

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ABSTRACT

Diffusion models have shown impressive capabilities in the generative AI space. These models have the capability to create images in a variety of styles from photorealistic and futuristic to many more artistic styles by simply using text prompts. This tutorial aims to introduce the underlying mechanisms that make these models successful along with hands-on exercises. The tutorial will start with explaining the diffusion concept with forward and reverse processes. Then, it will cover the fine-tuning process and the control procedures such as guidance and conditioning. The provided hands-on exercises will help apply these concepts on some real-world problems.

CCS CONCEPTS

• Applied computing • Information systems

KEYWORDS

diffusion models, diffusion, computer vision

ACM Reference format:

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1 INTRODUCTION

Generative AI is an exciting topic in machine learning (ML) where large-scale models are used to generate novel content. In the image domain, diffusion-based models have been successful in generating images with the help of text inputs. This tutorial aims to introduce the diffusion models and their applications in a hands-on fashion.

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The tutorial will start with explaining the diffusion concept with forward and reverse processes. Then, it will cover the fine-tuning process and the control procedures such as guidance and conditioning. There will be hands-on exercises to build the practical aspects of the mentioned methods.

2 TUTORIAL DESCRIPTION

Overall, this hands-on tutorial aims to introduce the diffusion concept and cover some methods to fine-tune and control the process such as guidance and conditioning.

2.1 Format

Half-day including the presentation, breaks and Q/A discussions

2.2 Target Audience

Research scientists and practitioners with introductory to medium level of understanding of some deep learning concepts such as convolutional neural networks and training methods would find this tutorial interesting.

2.3 Tutorial Outline

Outline of the tutorial is the following. Each section includes practical code examples.

- **Introduction to the diffusion:** The motivation, example use cases and the forward/reverse diffusion concepts [1, 2] are explained. The main architecture (U-Net) [3] and text encoders are introduced.
- **Fine-tuning, guidance and conditioning:** Finetuning is an important concept in machine learning to adapt pre-trained models to different tasks. Textual inversion [4], Dreambooth [5] and Low-Rank Adaptation of Large Language Models (LoRA) [6] are covered under the fine-tuning section. Tutorial extends on this discussion with guidance methods where the generation process is evaluated against a guidance function and conditioning methods where the available extra information such as labels and captions are also used in the process.
- **Latent diffusion and stable diffusion:** Diffusion concept is extended to include latent/stable diffusion [7] concepts.

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